

Enterprise Network Security Report

Simulated Secure Banking Infrastructure

VLAN Segmentation – ACL Enforcement – SSH Security – Syslog
Monitoring

Prepared by:

Saad Almutairi

Hamad Al-Hamdan

Abdulrahman Haddadi

Mohammed Alsaadoun

Nasser Al Haider

Imam Mohammad Ibn Saud Islamic University
Cybersecurity Diploma

Course:
NET-0216

May 2026

Section 1: Introduction

This project presents the design and implementation of a simulated secure enterprise banking network using Cisco Packet Tracer. The objective of the project was to build a segmented and security-focused network infrastructure capable of protecting sensitive internal resources from unauthorized access attempts.

The network was designed using VLAN segmentation, inter-VLAN routing, Access Control Lists (ACLs), SSH secure remote administration, and Syslog monitoring services. Security policies were implemented to isolate departments, secure administrative access, and protect internal web services from unauthorized users.

Additionally, attack simulation scenarios were performed to validate the effectiveness of the implemented security controls. The project demonstrates fundamental enterprise network security concepts and highlights the importance of layered security architecture in modern organizations.

Section 2: Network Architecture

The network topology was designed to simulate a secure enterprise banking environment consisting of multiple departments, internal services, and centralized security controls.

The infrastructure includes an edge router, core and access switches, multiple departmental VLANs, and dedicated internal servers. VLAN segmentation was implemented to logically separate departments and reduce unauthorized access between network segments.

The network contains the following VLANs:

- VLAN 10 – IT Department
- VLAN 20 – Finance Department
- VLAN 30 – Human Resources Department
- VLAN 40 – Guest Network
- VLAN 50 – Internal Servers

Inter-VLAN routing was configured using the Router-on-a-Stick technique on the edge router to enable controlled communication between VLANs.

Additionally, the infrastructure includes:

- A Web Server representing an internal banking service
- A Syslog Server used for centralized event monitoring and logging
- SSH secure remote administration for protected network management

Figure 1 shows the complete enterprise network topology implemented in Cisco Packet Tracer.

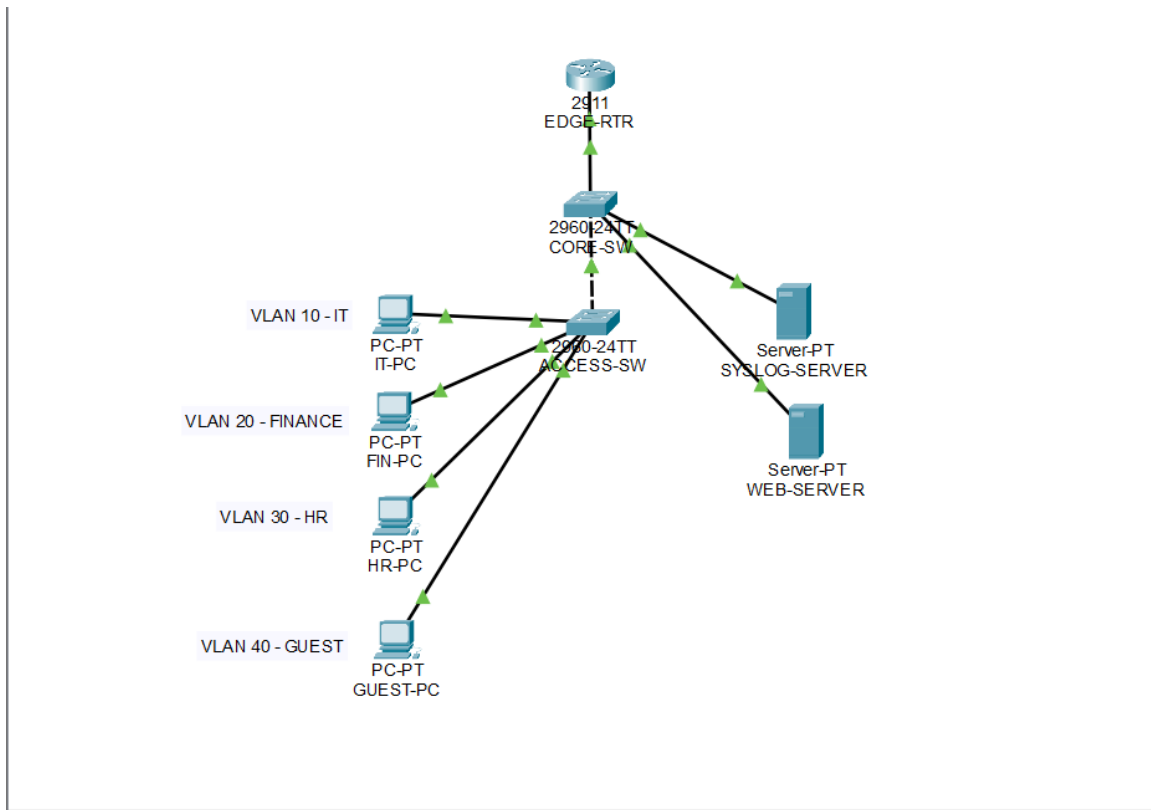


Figure 1: Enterprise Network Topology Implemented in Cisco Packet Tracer

Addressing Table

Device	IP Address	VLAN
IT-PC	192.168.10.10	VLAN 10
FIN-PC	192.168.20.10	VLAN 20
HR-PC	192.168.30.10	VLAN 30
GUEST-PC	192.168.40.10	VLAN 40
WEB-SERVER	192.168.50.20	VLAN 50
SYSLOG-SERVER	192.168.50.10	VLAN 50
EDGE-RTR G0/0.10	192.168.10.1	VLAN 10
EDGE-RTR G0/0.20	192.168.20.1	VLAN 20
EDGE-RTR G0/0.30	192.168.30.1	VLAN 30
EDGE-RTR G0/0.40	192.168.40.1	VLAN 40
EDGE-RTR G0/0.50	192.168.50.1	VLAN 50

Section 3: Security Implementation

Multiple security mechanisms were implemented within the network infrastructure to enhance confidentiality, access control, administrative security, and network monitoring.

The first security layer was VLAN segmentation, which logically separated departments into isolated broadcast domains. This approach reduced unnecessary communication between departments and improved internal network security.

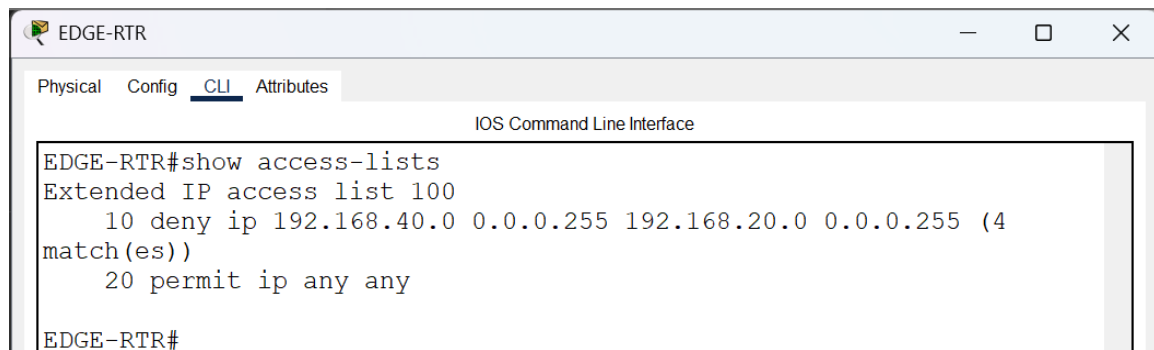
Access Control Lists (ACLs) were configured on the edge router to restrict unauthorized communication between VLANs. Specifically, devices connected to the Guest VLAN were prevented from accessing sensitive internal resources such as the Finance network and internal servers.

To secure remote administration, SSH version 2 was configured on the edge router. RSA encryption keys and local

authentication credentials were implemented to protect administrative access and prevent insecure remote management methods.

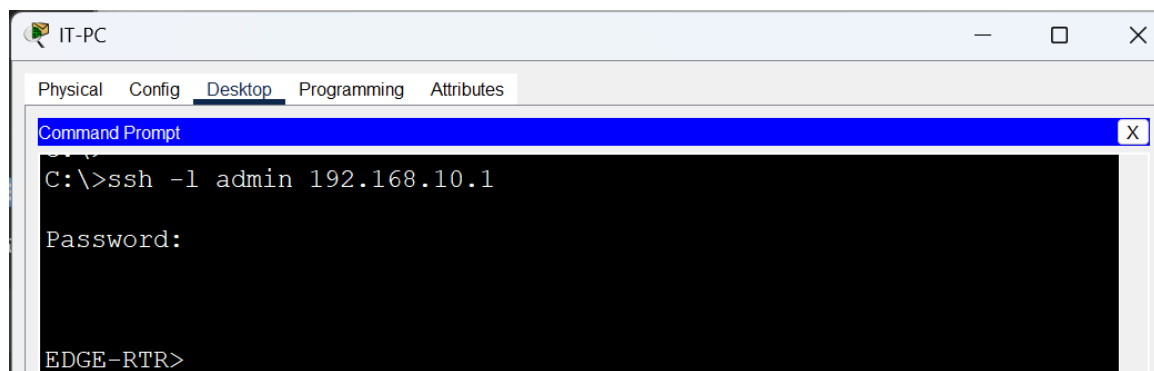
Additionally, a centralized Syslog monitoring server was deployed to record and monitor network events and configuration activities. This allowed administrative actions and security-related events to be logged and reviewed from a centralized location.

The implemented security controls provided a layered security architecture that improved network protection and reduced the risk of unauthorized access.



```
EDGE-RTR#show access-lists
Extended IP access list 100
 10 deny ip 192.168.40.0 0.0.0.255 192.168.20.0 0.0.0.255 (4
match(es))
 20 permit ip any any
EDGE-RTR#
```

Figure 2: ACL Configuration for Restricting Unauthorized VLAN Access



```
C:\>ssh -l admin 192.168.10.1
Password:
EDGE-RTR>
```

Figure 3: Secure Remote Administration Using SSH

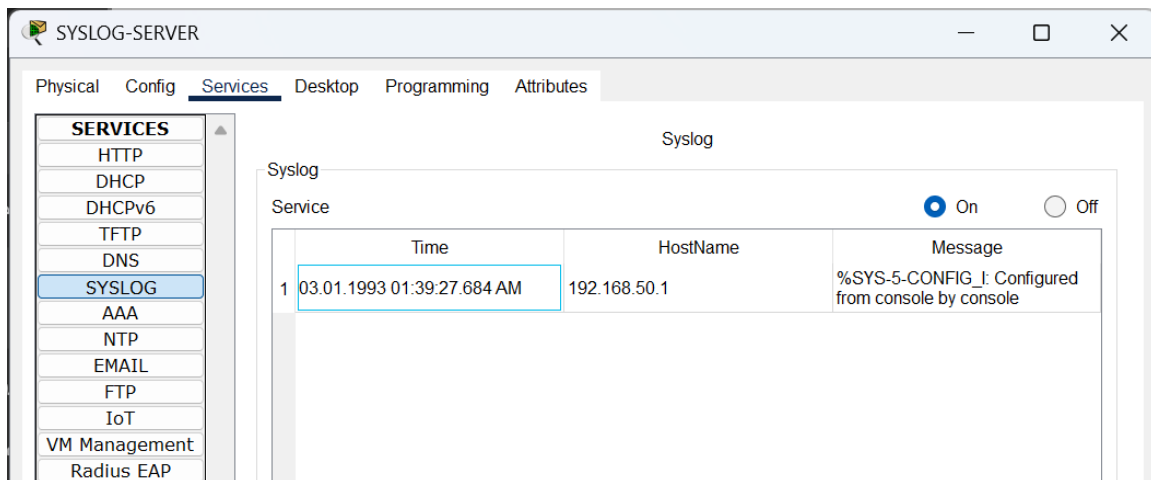


Figure 4: Centralized Syslog Monitoring and Event Logging

Section 4: Attack Simulation and Security Validation

Attack simulation scenarios were performed to evaluate the effectiveness of the implemented security controls within the enterprise network.

The primary simulation involved an unauthorized access attempt originating from the Guest VLAN. The objective of the attack was to access sensitive resources located inside the Finance VLAN and the internal banking web server.

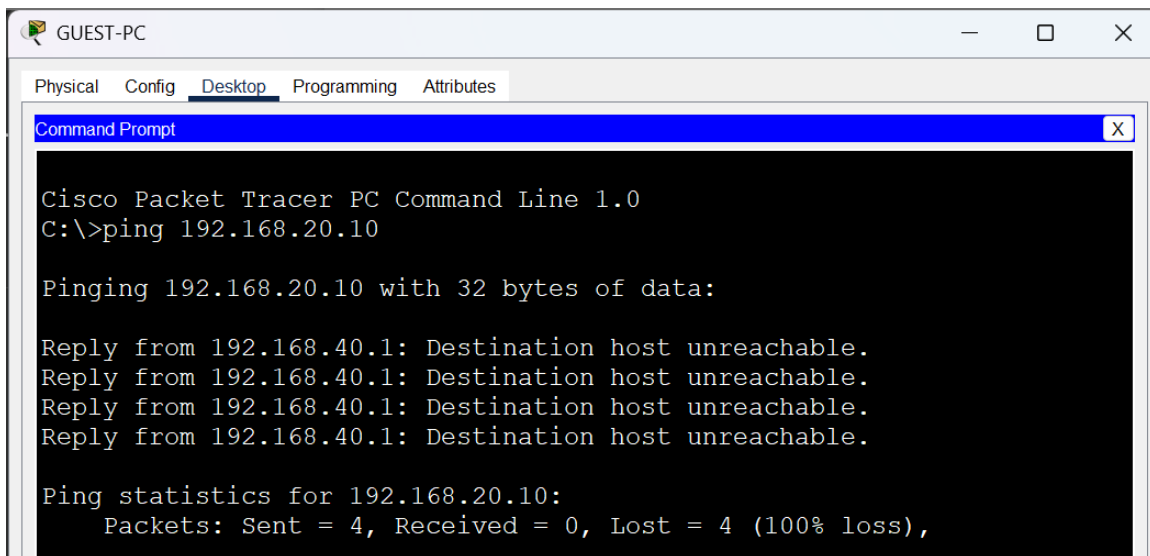
Access Control Lists (ACLs) successfully prevented the Guest network from communicating with protected internal resources. When the Guest device attempted to access the Finance VLAN and internal web services, the communication requests were denied according to the configured security policies.

In contrast, authorized devices located within trusted VLANs such as the IT department were able to access internal resources successfully. This validated that the implemented

security mechanisms enforced proper access restrictions while maintaining legitimate communication.

Additional validation was performed through SSH secure administration testing and Syslog event monitoring. Administrative activities and configuration changes were successfully logged and monitored through the centralized Syslog server.

The attack simulation demonstrated the effectiveness of VLAN segmentation, ACL enforcement, secure remote administration, and centralized monitoring in protecting enterprise network resources from unauthorized access attempts.



The screenshot shows a window titled 'GUEST-PC' with tabs for 'Physical', 'Config', 'Desktop', 'Programming', and 'Attributes'. The 'Desktop' tab is active, displaying a 'Command Prompt' window. The command prompt shows the following text:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Reply from 192.168.40.1: Destination host unreachable.
Reply from 192.168.40.1: Destination host unreachable.
Reply from 192.168.40.1: Destination host unreachable.
Reply from 192.168.40.1: Destination host unreachable.

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Figure 5: Unauthorized Access Attempt from Guest VLAN Blocked by ACL Policy

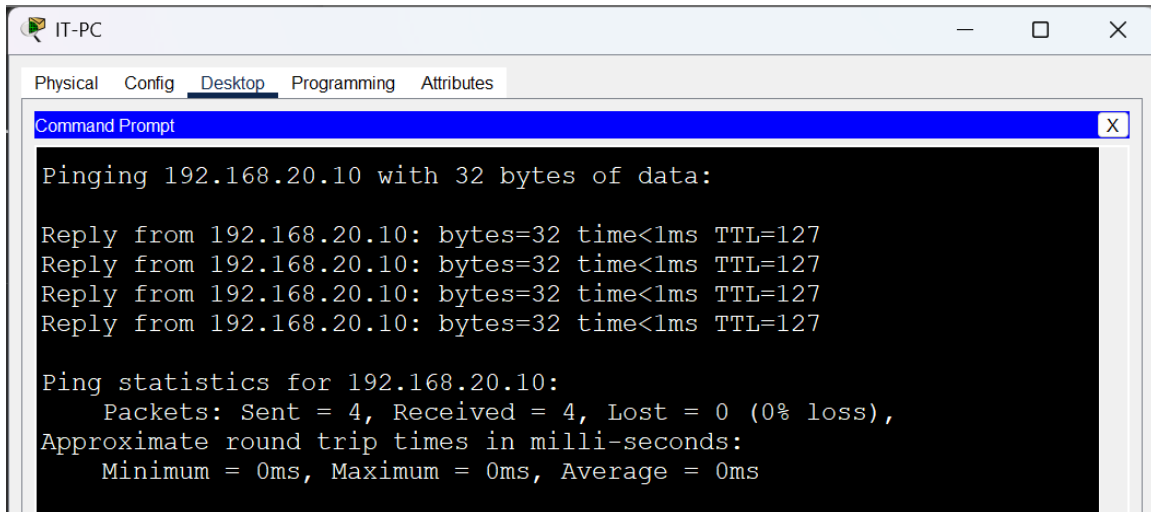


Figure 6: Authorized Communication Between Trusted VLANs

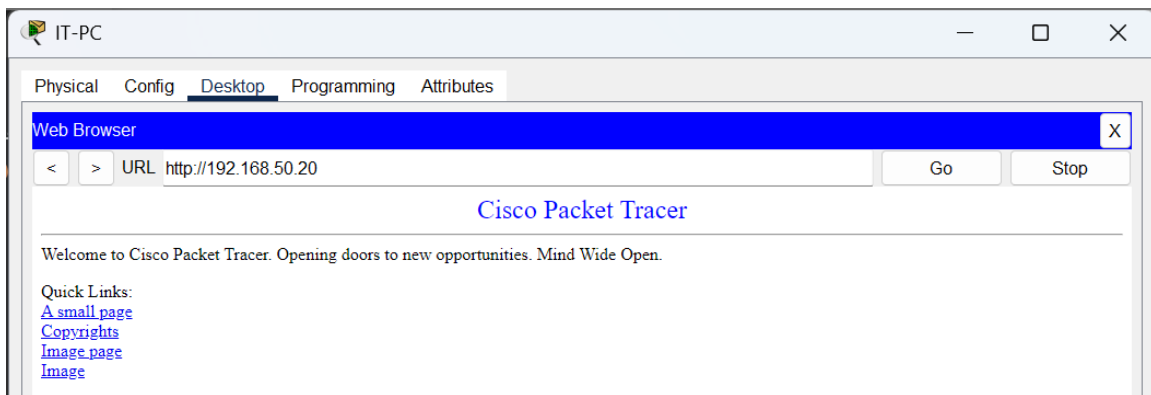


Figure 7: Authorized Access to Internal Banking Web Service

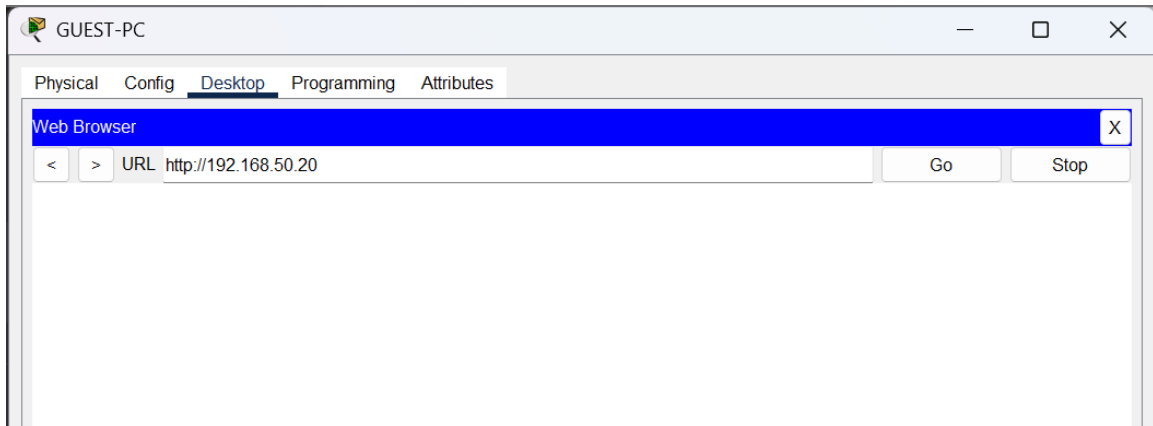


Figure 8: Unauthorized Access to Internal Web Service Successfully Blocked

Section 5: Testing and Results

Multiple connectivity and security tests were performed to verify the functionality and effectiveness of the implemented network security mechanisms.

Initial testing confirmed successful communication between authorized VLANs using inter-VLAN routing. Devices located in trusted departments were able to communicate with internal resources and access the internal banking web server successfully.

Security validation tests confirmed that unauthorized communication attempts from the Guest VLAN were successfully blocked by the configured ACL policies. This demonstrated that sensitive resources such as the Finance department and internal servers were protected against unauthorized access.

SSH remote administration testing verified that secure encrypted remote management was functioning correctly. Administrative access was successfully authenticated using local credentials and encrypted communication channels.

Additionally, Syslog monitoring successfully recorded administrative and system events, confirming that centralized monitoring and logging services were operating correctly.

Overall, the implemented security controls operated successfully and achieved the intended protection objectives within the simulated enterprise network environment.

Section 6: Conclusion

This project demonstrated the practical implementation of a secure enterprise network using Cisco Packet Tracer.

The network infrastructure was designed using VLAN segmentation, inter-VLAN routing, Access Control Lists (ACLs), SSH secure administration, and centralized Syslog monitoring to simulate a realistic banking network environment.

Through attack simulation scenarios, the project successfully demonstrated how security controls can prevent unauthorized access to sensitive internal resources while maintaining legitimate communication between trusted departments.

The implementation highlighted the importance of layered security architecture, access control enforcement, secure remote management, and continuous monitoring in protecting enterprise networks.

Overall, the project provided practical experience in designing, implementing, testing, and validating network security mechanisms within a simulated enterprise infrastructure.